Digital & Technological Solutions

Introduction & Evolution of Digital Systems

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Govt. Degree College Beerwah

SEMESTER – 1st to 3rd

VALUE ADDED COURSE

DTS022V DIGITAL AND TECHNOLOGICAL SOLUTIONS

CREDITS: 02

Course Objectives:

- *To gain familiarity with digital paradigms*
- *To sensitize about role & significance of digital technology*
- *To provide know how of communications & networks*
- *To bring awareness about the e-governance and Digital India initiatives*
- *To provide a flavour of emerging technologies - Cloud, Big Data, ai 3D printing*

Course Outcome:

1. *Knowledge about digital paradigm.*
2. *Realization of importance of digital technology, digital financial tools, e-commerce.*
3. *Know-how of communication and networks.*
4. *Familiarity with the e-governance and Digital India initiatives*
5. *An understanding of use & applications of digital technology.*
6. *Basic knowledge of all machine learning and big data*

COURSE CONTENTS:

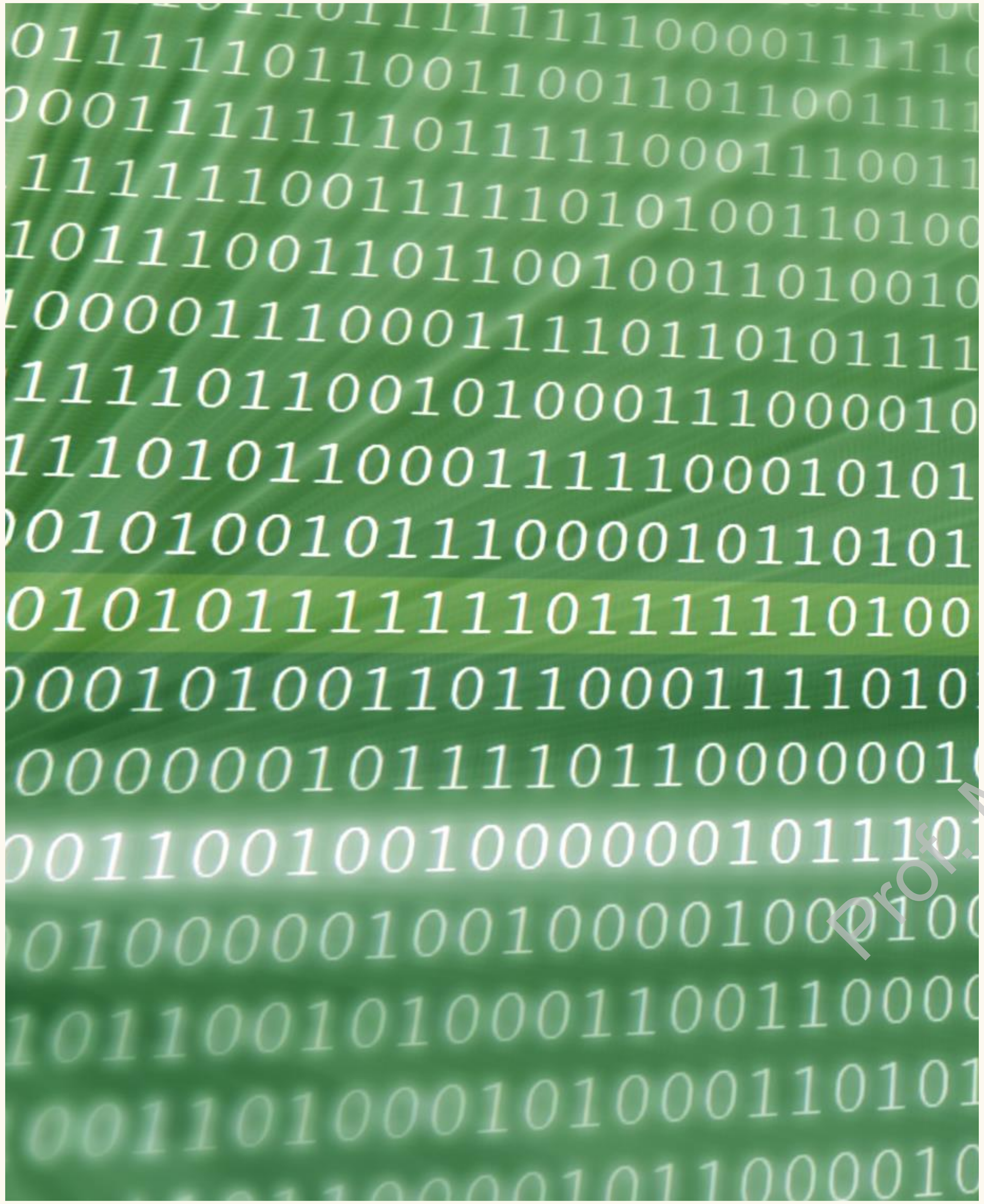
UNIT I

Introduction & Evolution of Digital Systems. Role & Significance of Digital Technology. Information & Communication Technology & Tools. Computer System & it's working, Software and its types. Operating Systems: Types and Functions. Problem Solving: Algorithms and Flowcharts.

Communication Systems: Principles, Model & Transmission Media. Computer Networks & Internet: Concepts & Applications, WWW, Web Browsers, Search Engines, Messaging, Email, Social Networking. Computer Based Information System: Significance & Types. E-commerce & Digital Marketing: Basic Concepts, Benefits & Challenges.

UNIT II

Digital India & e-Governance: Initiatives, Infrastructure, Services and Empowerment. Digital Financial Tools: Unified Payment Interface, Aadhar Enabled Payment System, USSD, Credit / Debit Cards, e-Wallets,



What are digital systems?

Digital systems are electronic devices that use binary digits to store, process and transmit information.

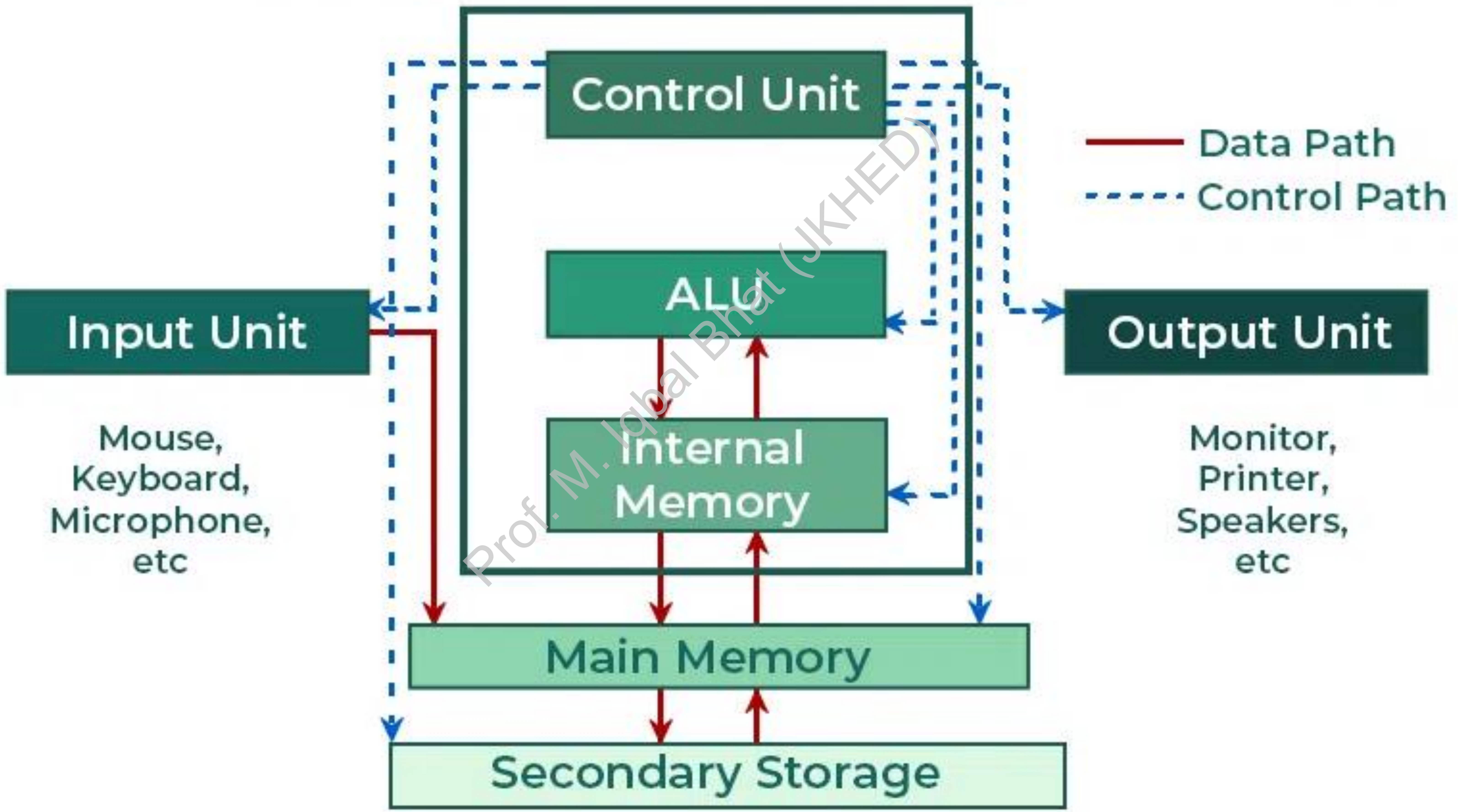
They have evolved from simple calculators to complex devices such as computers, smartphones and tablets.

Digital systems have revolutionized various fields including communication, entertainment, healthcare and education.

Key Components of Digital Systems

- **Input devices:** Input devices are used to enter data into a digital system. Common input devices include keyboards, mice, scanners, and cameras.
- **Processing units**: Processing units are the brains of digital systems. They perform arithmetic and logical operations on digital signals to perform various tasks. Common processing units include central processing units (CPUs) and graphics processing units (GPUs).
- **Memory**: Memory is used to store data and program instructions. Common types of memory include random access memory (RAM) and read-only memory (ROM).
- **Output devices:** Output devices are used to display or output the results of the processing that has been performed on the data. Common output devices include monitors, printers, and speakers.
- **Communication channels**: Communication channels are used to connect digital systems to each other and to other devices. Common communication channels include Ethernet cables, Wi-Fi networks, and cellular networks.

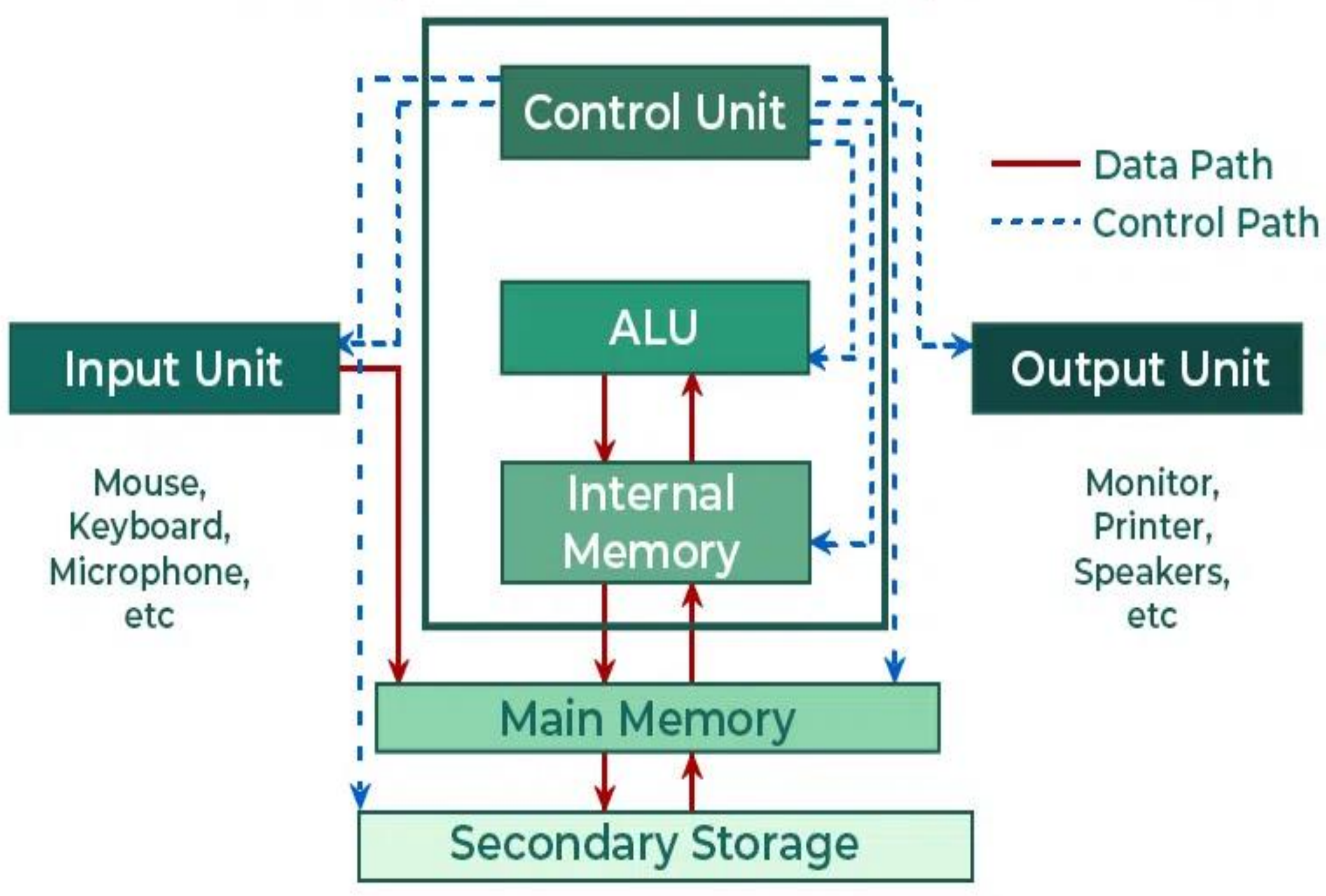
Key Components of Digital Systems



Input Unit

An input unit of a computer system performs the following functions:

1. It accepts (or reads) instructions and data from outside world
2. It converts these instructions and data in computer acceptable form
3. It supplies the converted instructions and data to the computer system for further processing



Keyboard

Mouse

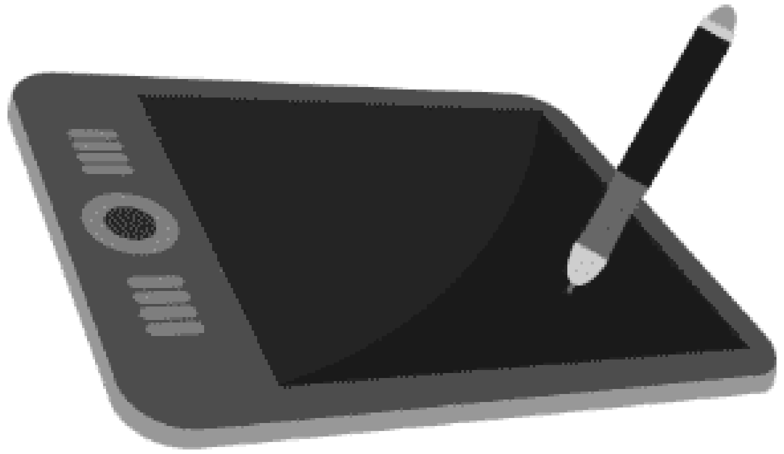
Joystick

Touchpad

Stylus

Trackball

Scanner



Graphic Tablet

Touchscreen

Microphone

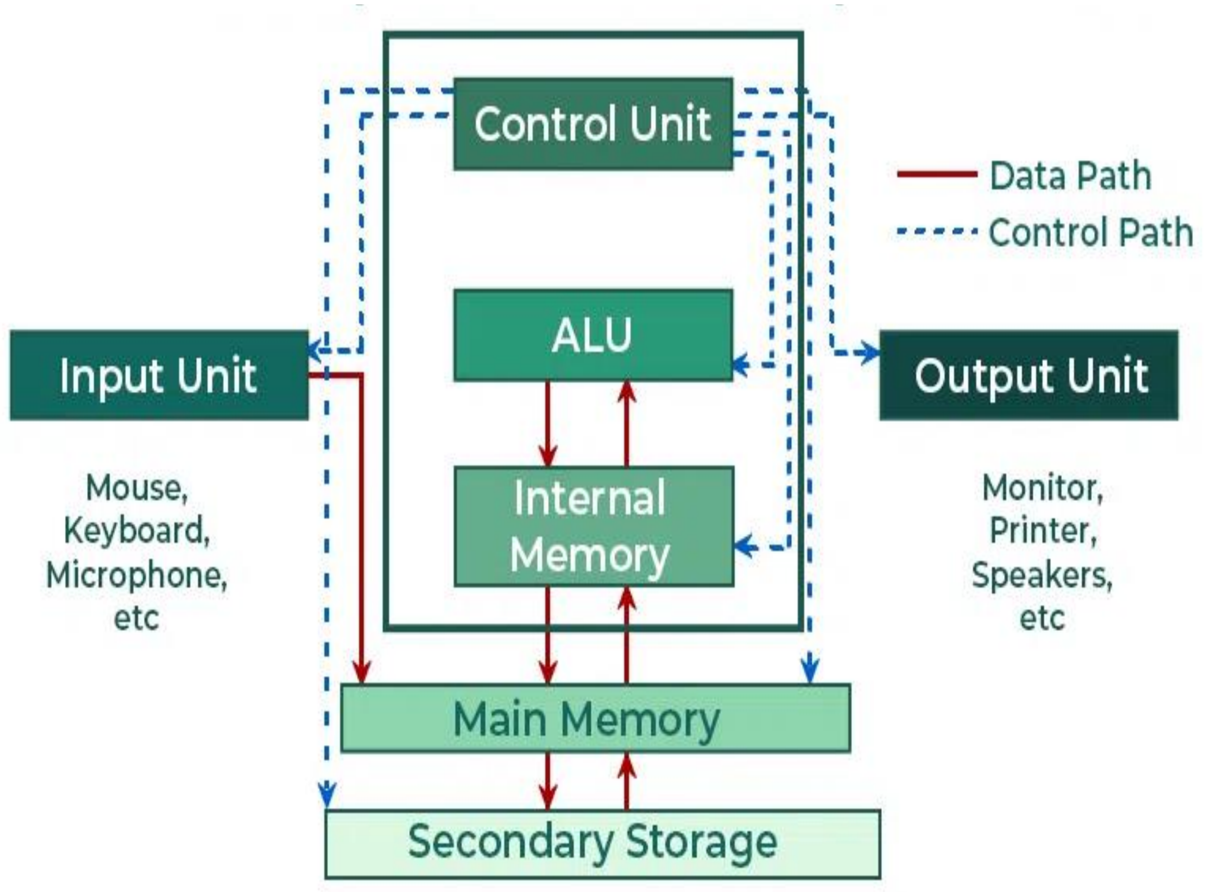
Camera

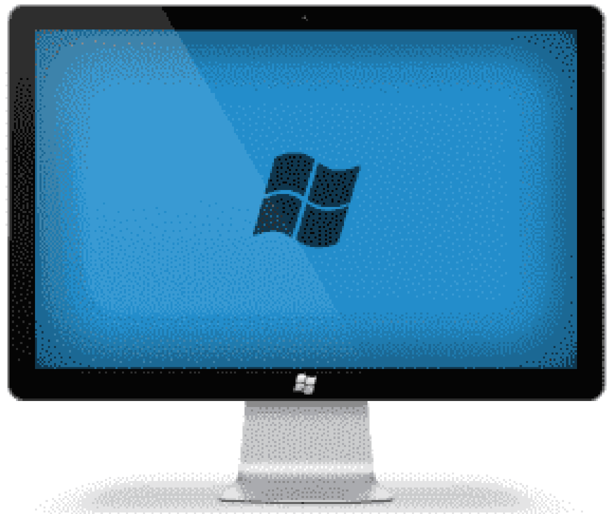
Webcam

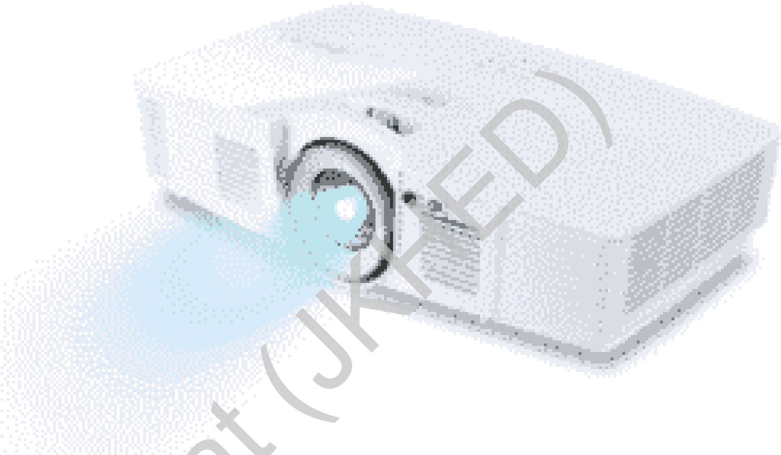
Output Unit

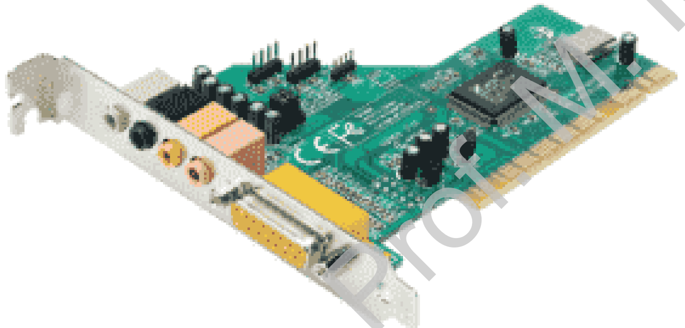
An output unit of a computer system performs the following functions:

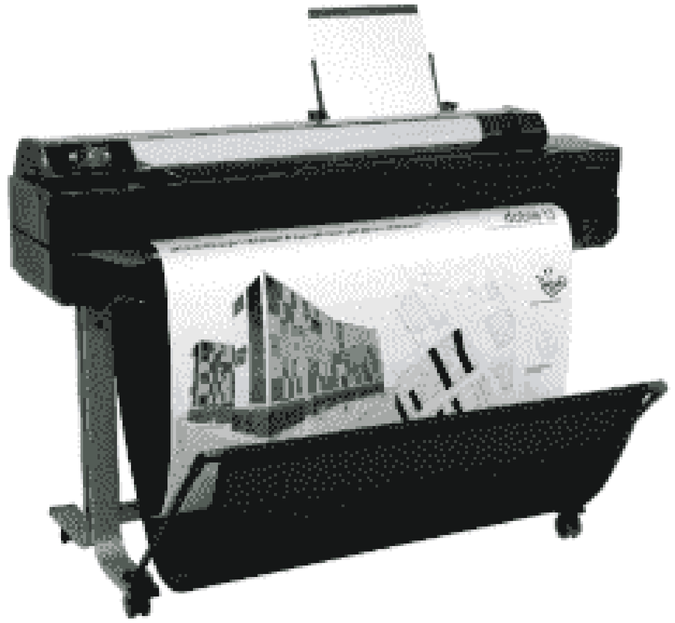
1. It accepts the results produced by the computer, which are in coded form and hence, cannot be easily understood by us
2. It converts these coded results to human acceptable (readable) form
3. It supplies the converted results to outside world







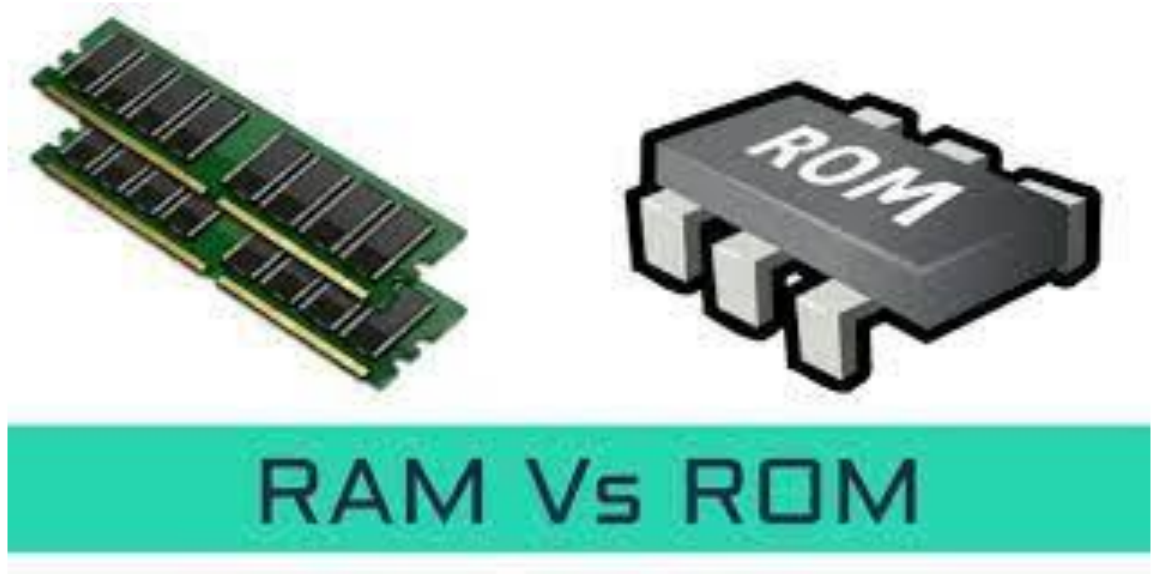



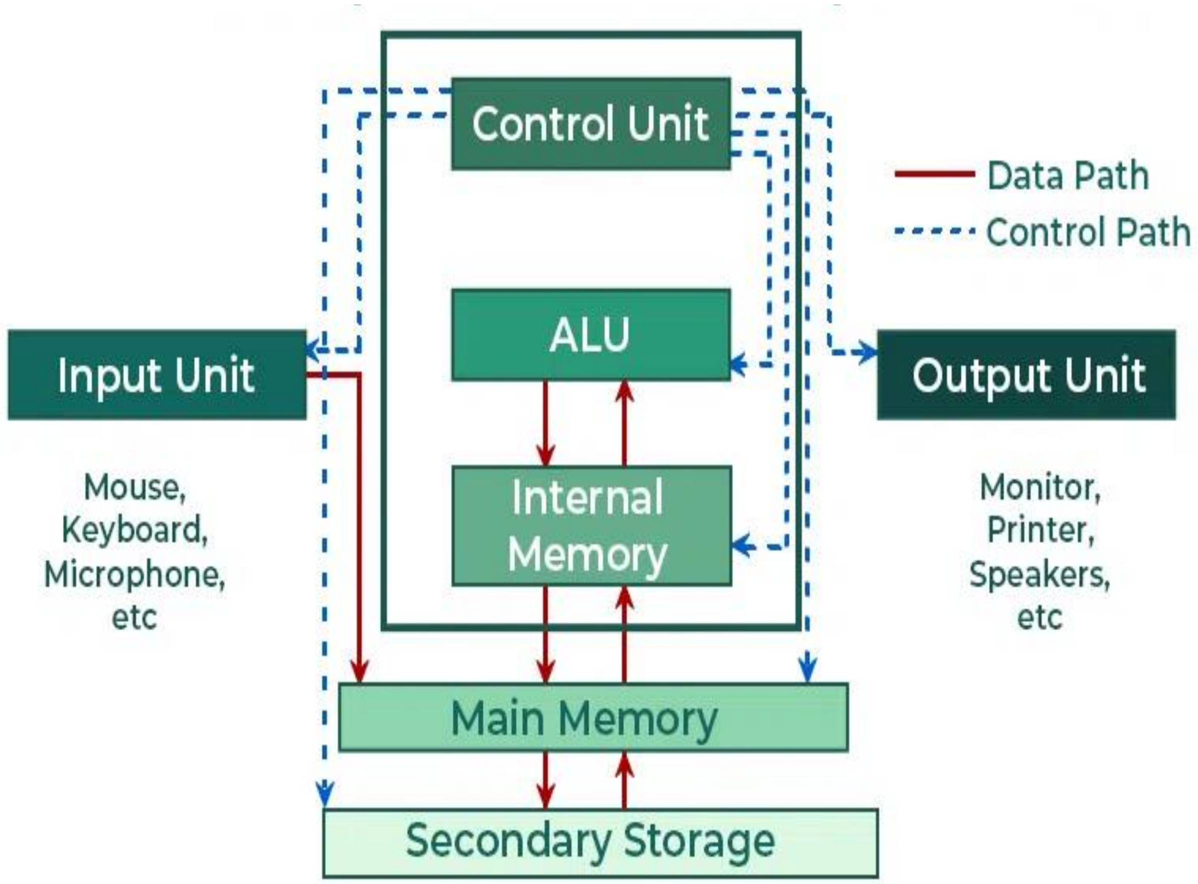


Storage Unit

The storage unit of a computer system holds (or stores) the following :

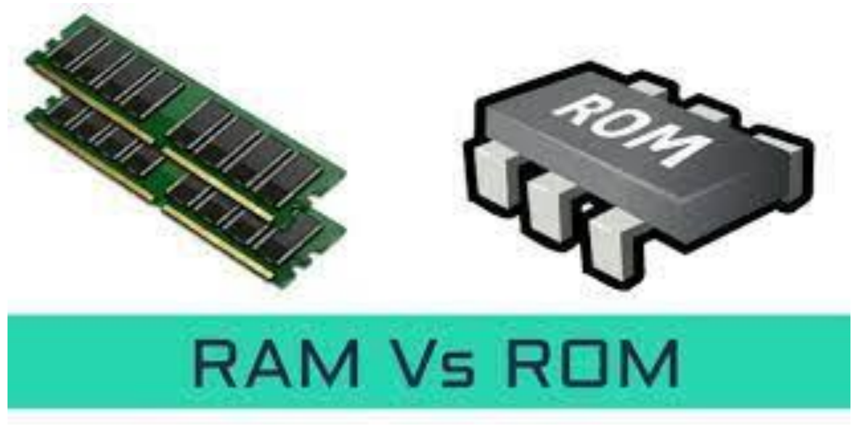
1. Data and instructions required for processing (received from input devices)
2. Intermediate results of processing
3. Final results of processing, before they are released to an output device





Two Types of Storage

- **Primary storage**
  - Used to hold running program instructions
  - Used to hold data, intermediate results, and results of ongoing processing of job(s)
  - Fast in operation
  - Small Capacity
  - Expensive
  - Volatile (looses data on power dissipation)



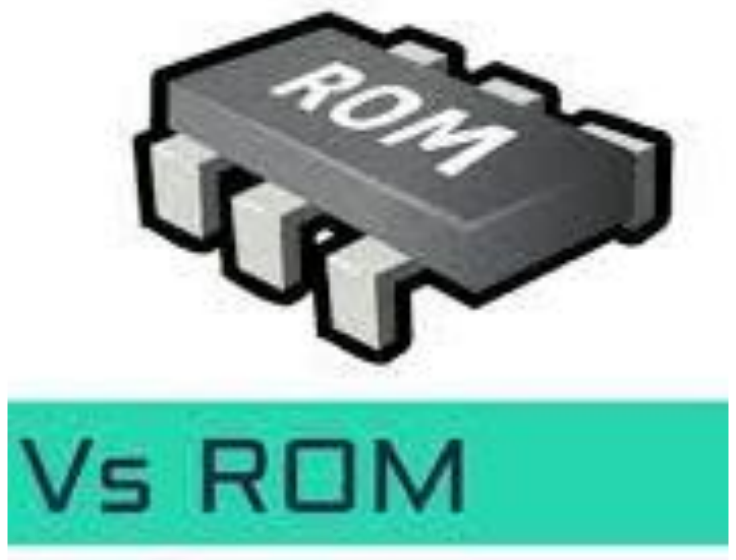
- **Secondary storage**
  - Used to hold stored program instructions
  - Used to hold data and information of stored jobs
  - Slower than primary storage
  - Large Capacity
  - Lot cheaper that primary storage
  - Retains data even without power

RAM (Random Access Memory)

Random Access Memory (RAM) –

- It is also called read-write *memory* or the *main memory* or the *primary memory*.
- The programs and data that the CPU requires during the execution of a program are stored in this memory.
- It is a volatile memory as the data is lost when the power is turned off.
- RAM is further classified into two types:
  - *SRAM (Static Random Access Memory)* and
  - *DRAM (Dynamic Random Access Memory).*

ROM (Read Only Memory)



Read Only Memory (ROM) –

- Stores crucial information essential to operate the system, like the program essential to boot the computer.
- It is not volatile (nonvolatile).
- Always retains its data.
- Used in embedded systems or where the programming needs no change.
- Used in calculators and peripheral devices.
- ROM is further classified into four types-
    - MROM, PROM, EPROM, and EEPROM.

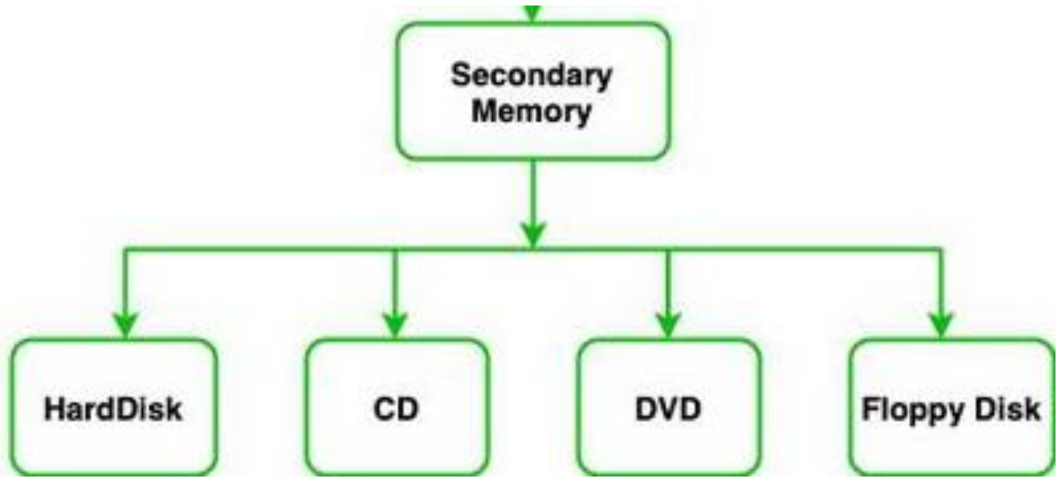
Difference	RAM	ROM
Data retention	RAM is a volatile memory which could store the data as long as the power is supplied.	ROM is a non-volatile memory which could retain the data even when power is turned off.
Working type	Data stored in RAM can be retrieved and altered.	Data stored in ROM can only be read.
Use	Used to store the data that has to be currently processed by CPU temporarily.	It stores the instructions required during bootstrap of the computer.
Speed	It is a high-speed memory.	It is much slower than the RAM.
CPU Interaction	The CPU can access the data stored on it.	The CPU can not access the data stored on it unless the data is stored in RAM.
Size and Capacity	Small size with less capacity.	Large size with higher capacity.
Used as/in	CPU Cache, Primary memory.	Firmware, Micro-controllers
Accessibility	The data stored is easily accessible	The data stored is not as easily accessible as in RAM
Cost	Costlier	cheaper than RAM.

Secondary Storage:

- Secondary memory is used for different purposes but the main purposes of using secondary memory are:
  - **Permanent storage:** As we know that primary memory stores data only when the power supply is on, it loses data when the power is off. So we need a secondary memory to store data permanently even if the power supply is off.
  - **Large Storage:** Secondary memory provides large storage space so that we can store large data like videos, images, audio, files, etc permanently.
  - **Portable:** Some secondary devices are removable. So, we can easily store or transfer data from one computer or device to another.

Secondary Storage:

- The secondary memory stores data and instructions permanently.
- The information can be stored in secondary memory for a long time (years), and is generally permanent in nature unless erased by the user.
- It is a non-volatile memory.
- It provides back-up storage for data and instructions.
- Hard disk drives, floppy drives, magnetic tape drives and optical disk drives are some examples of storage devices.
- The data and instructions that are currently not being used by CPU, but may be required later for processing, are stored in secondary memory.
- Secondary memory has a high storage capacity than primary memory.
- Secondary memory is also cheaper than primary memory.
- It takes a longer time to access the data and instructions stored in secondary memory than
- in primary memory.

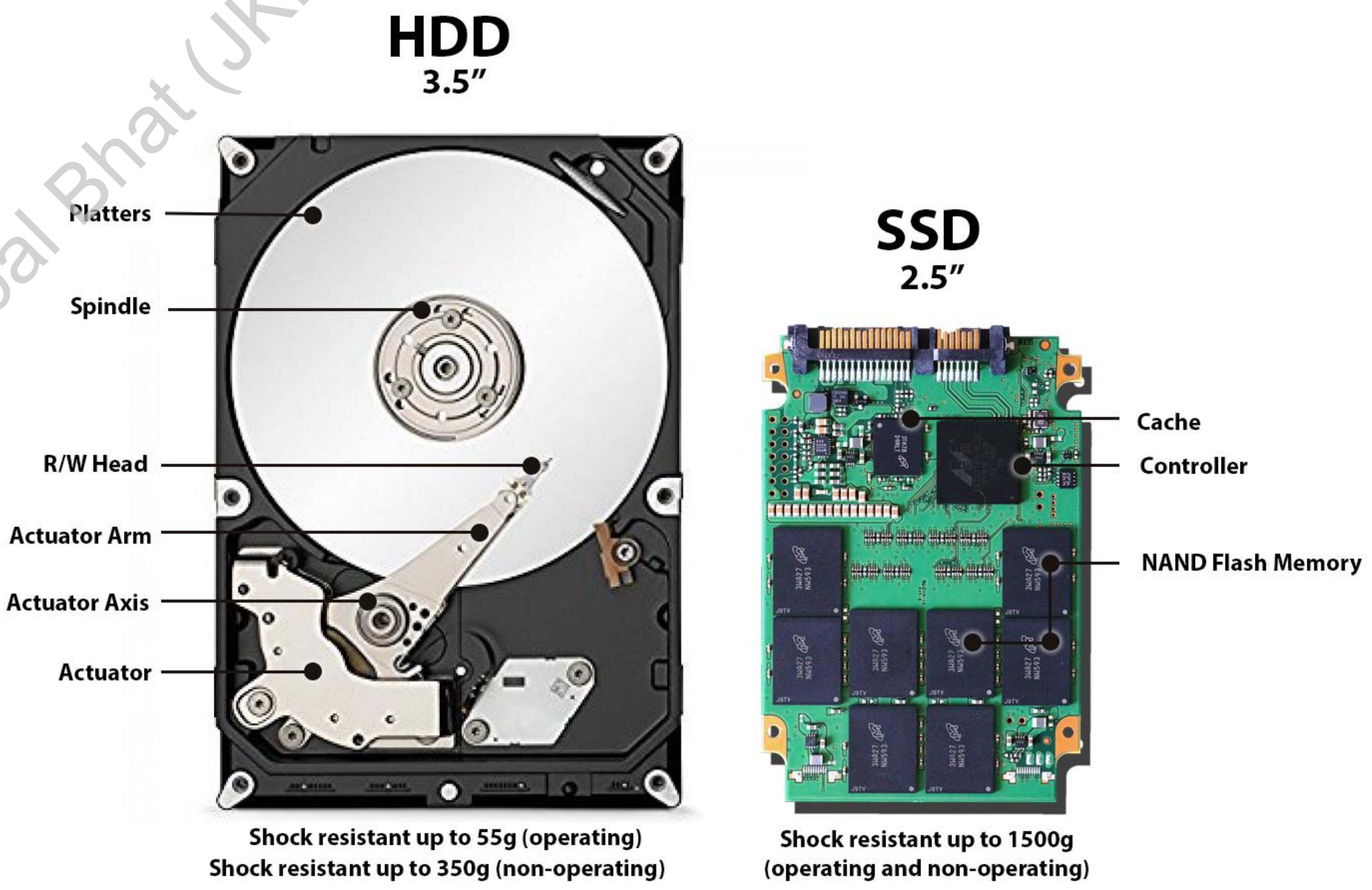


Secondary Storage Memory:

- Types of Secondary storage:
  - Fixed
  - Removable

Fixed Storage

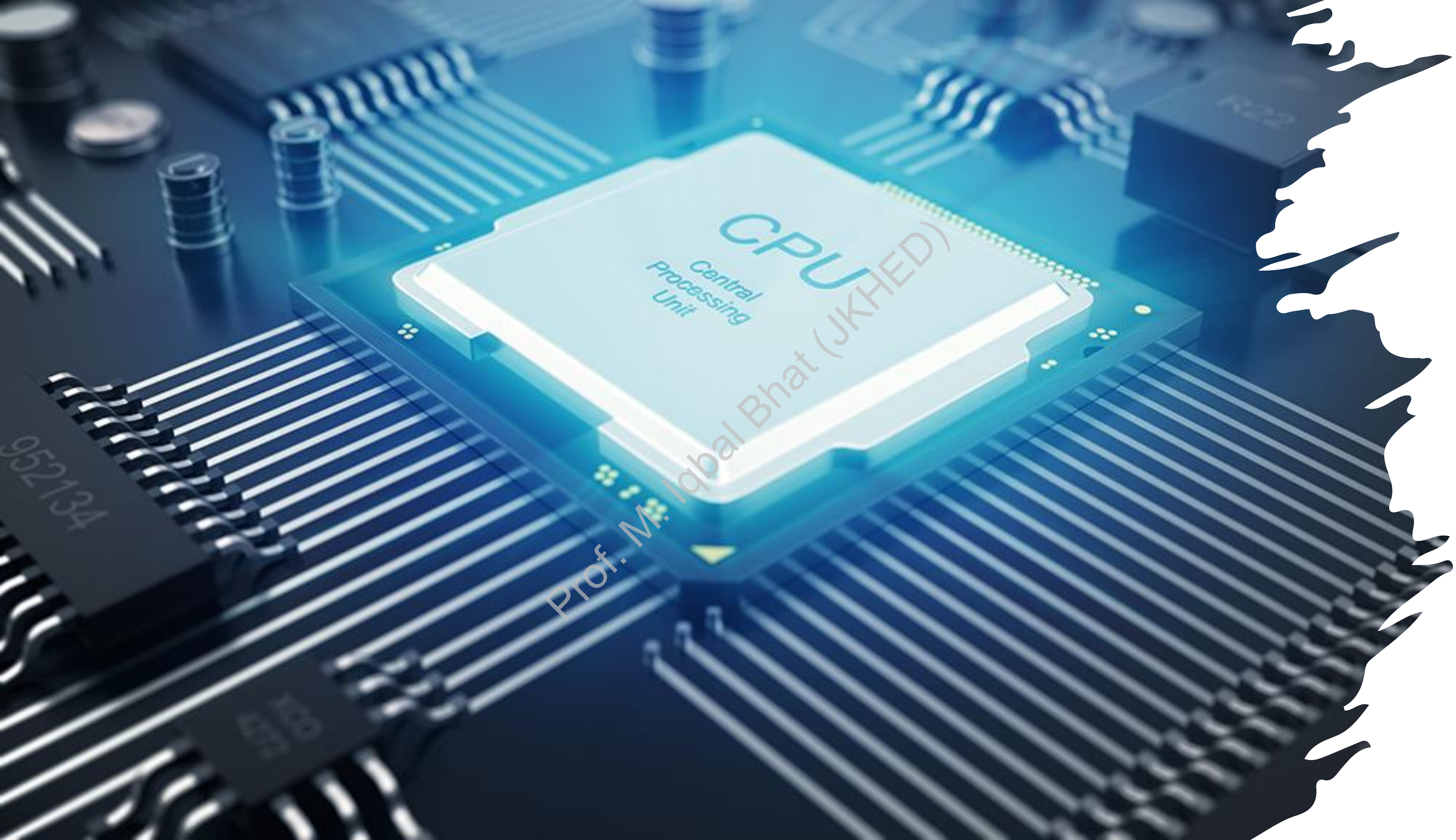
- In secondary memory, a fixed storage is an internal media device that is used to store data in a computer system. Fixed storage is generally known as fixed disk drives or hard drives:
  - **Internal flash memory (rare)**
  - **SSD (solid-state disk)**
  - **Hard disk drives (HDD)**



Removable Storage

- In secondary memory, removable storage is an external media device that is used to store data in a computer system. Removable storage is generally known as disks drives or external drives.:
  - **Optical discs (like CDs, DVDs, Blu-ray discs, etc.)**
  - **Memory cards**
  - **Floppy disks**
  - **Magnetic tapes**
  - **Disk packs**
  - **Paper storage**

(like punched tapes, punched cards, etc.)


CPU
Central
Processing
Unit
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Central Processing Unit (CPU)

- A Central Processing Unit is also called a processor, central processor, or microprocessor.
- It carries out all the important functions of a computer.
- It receives instructions from both the hardware and active software and produces output accordingly.
- It stores all important programs like operating systems and application software.
- CPU also helps Input and output devices to communicate with each other. Owing to these features of CPU, it is often referred to as the brain of the computer.

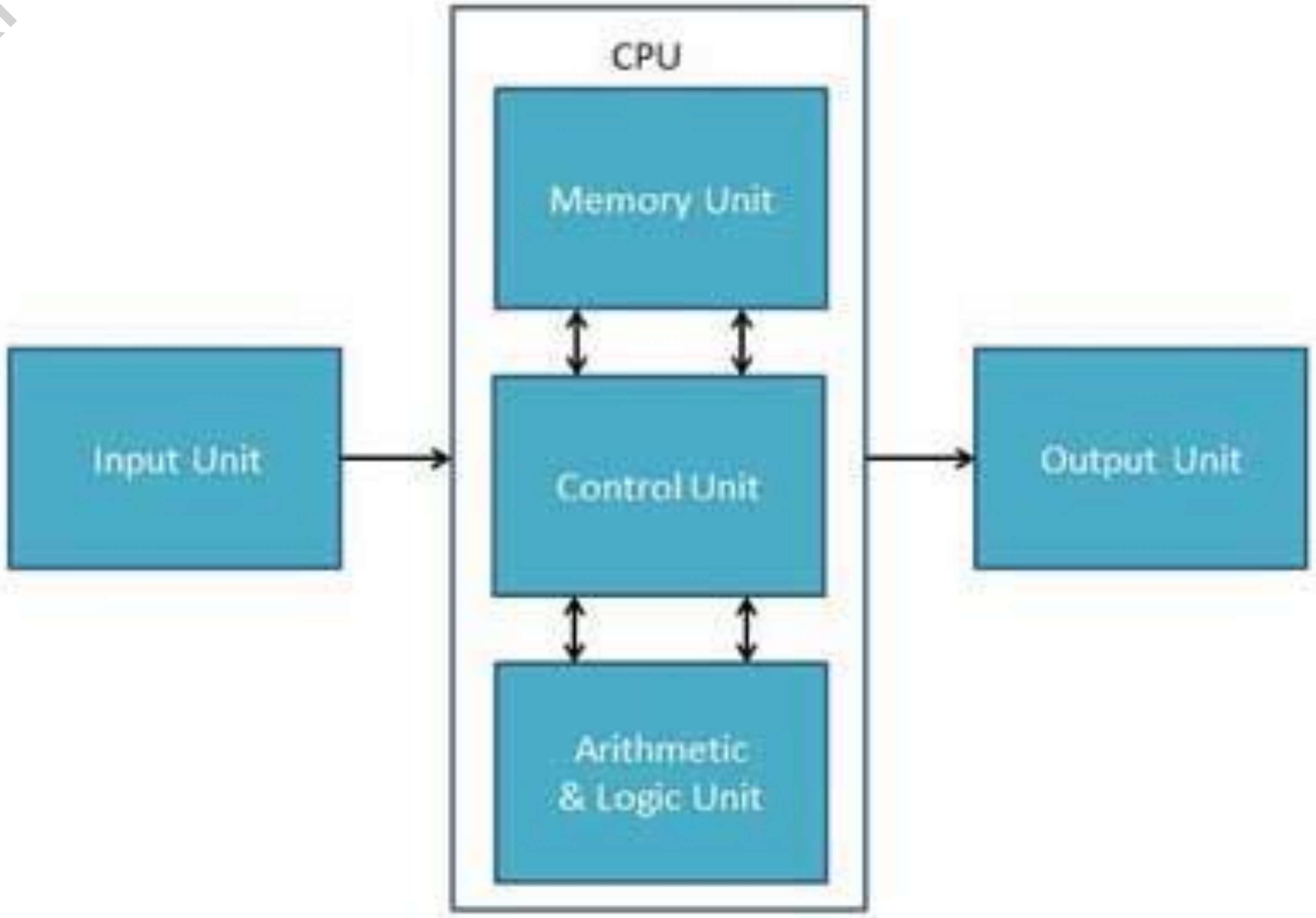
Arithmetic Logic Unit (ALU) + Control Unit (CU) = Central Processing Unit (CPU)

- It is the brain of a computer system
- It is responsible for controlling the operations of all other units of a computer system

Arithmetic Logic Unit (ALU)

Arithmetic Logic Unit of a computer system is the place where the actual executions of instructions takes place during processing operation

- **Logical Operations:** The logical operations consist of NOR, NOT, AND, NAND, OR, XOR, and more.
- **Bit-Shifting Operations:** It is responsible for displacement in the locations of the bits to the by right or left by a certain number of places that are known as a multiplication operation.
- **Arithmetic Operations:** Although it performs multiplication and division, this refers to bit addition and subtraction. But multiplication and division operations are more costly to make. In the place of multiplication, addition can be used as a substitute and subtraction for division.



Control Unit (CU)

Control Unit of a computer system manages and coordinates the operations of all other components of the computer system

- It is the circuitry in the control unit, which makes use of electrical signals to instruct the computer system for executing already stored instructions.
- It takes instructions from memory and then decodes and executes these instructions.
- So, it controls and coordinates the functioning of all parts of the computer.
- The Control Unit's main task is to maintain and regulate the flow of information across the processor. It does not take part in processing and storing data.

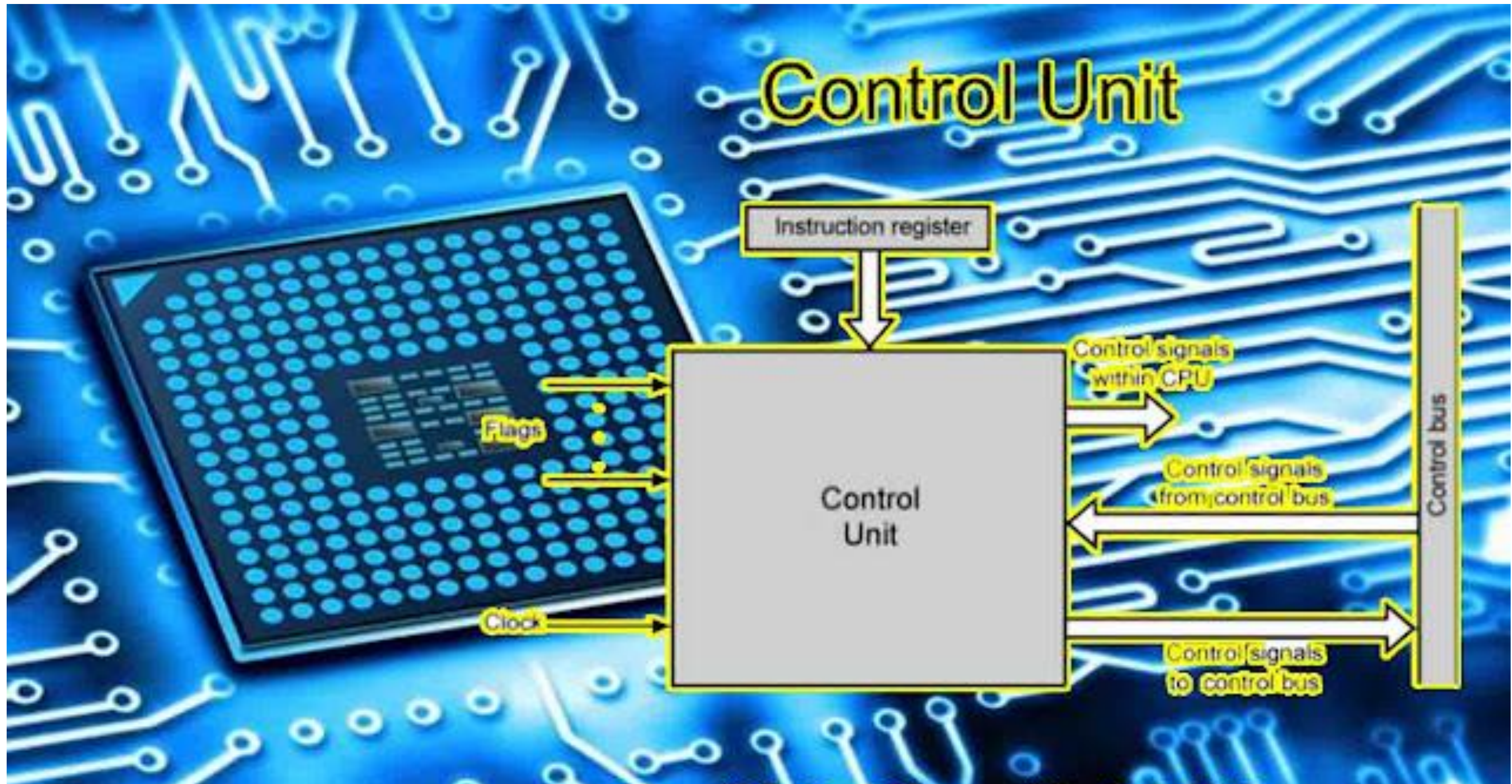


Figure 16.4 Block Diagram of the Control Unit

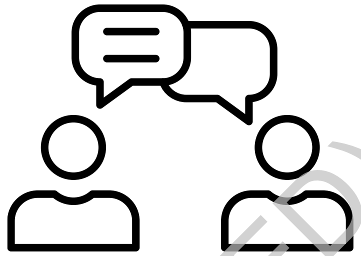
Role of digital technology

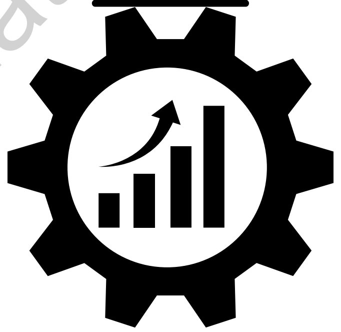
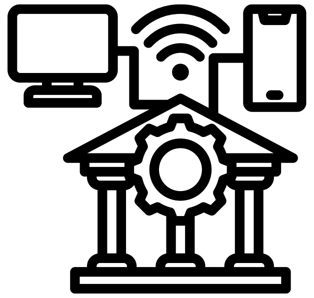
Digital technology plays a vital role in many aspects of our lives. It is used in:

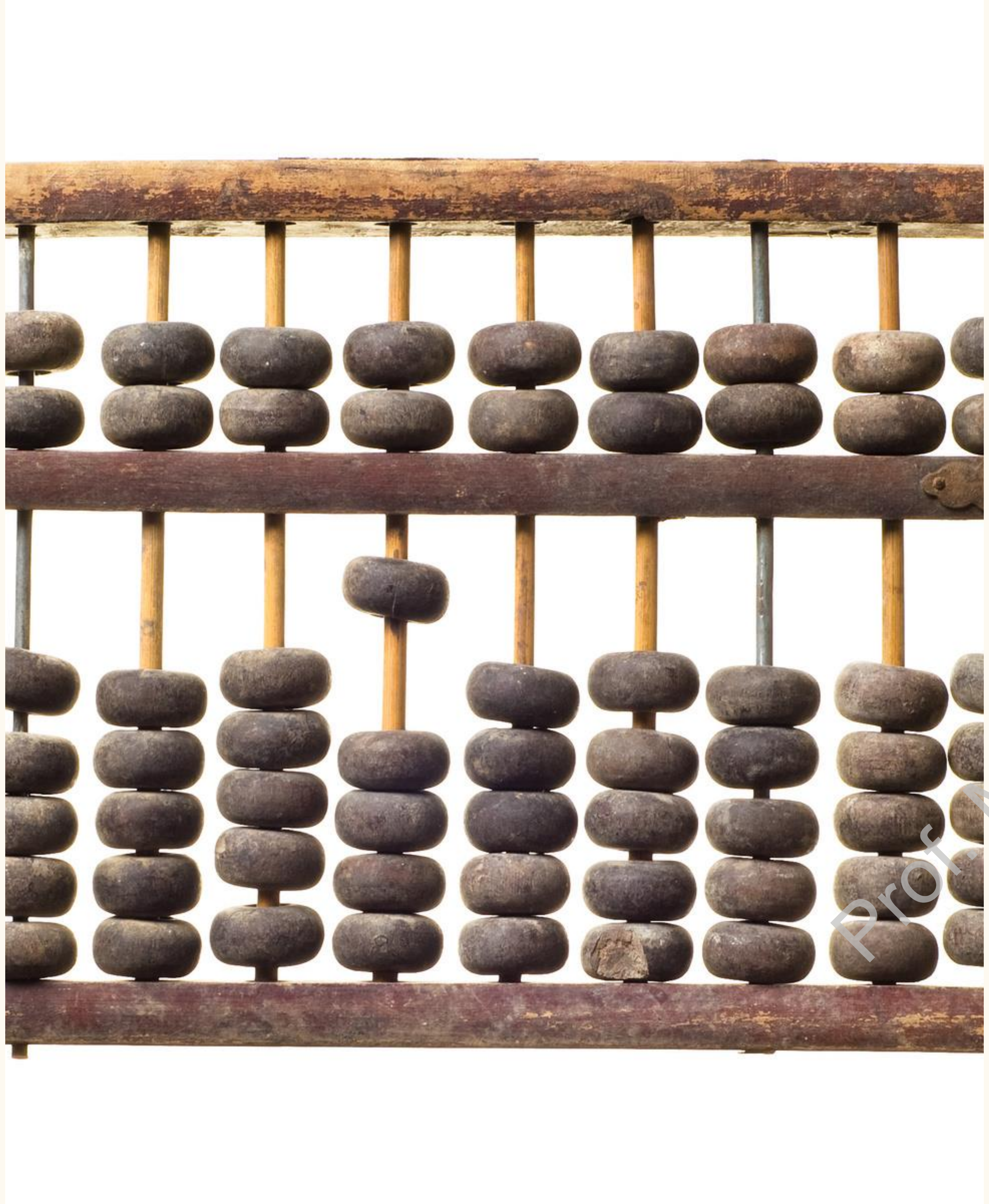
- **Communication:** Digital technology has made it easier and faster to communicate with people all over the world. We can use email, social media, and instant messaging to stay in touch with friends and family, and to collaborate with colleagues on projects.
- **Education:** Digital technology is being used to transform education. Students can use online resources to learn at their own pace and to access information from all over the world. Teachers can use digital tools to create engaging and interactive learning experiences.
- **Business:** Digital technology is essential for businesses of all sizes. Businesses use digital technology to manage their operations, to communicate with customers, and to sell their products and services online.
- **Healthcare:** Digital technology is being used to improve healthcare delivery and to develop new medical treatments and devices. For example, digital technology is being used to diagnose diseases, to perform surgery, and to monitor patients' health remotely.
- **Entertainment:** Digital technology has revolutionized the way we consume entertainment. We can use digital devices to watch movies and TV shows, to listen to music, and to play games.

Significance of digital technology

Digital technology is significant because it has the power to improve our lives in many ways. It can help us to:

- **Communicate more effectively** 
- **Learn new things more easily** 

- **Work more efficiently** 
- **Access healthcare services more easily** 
- **Enjoy entertainment more fully** 
- **impart effective and inclusive government** 



History of digital systems

Digital systems have been around for centuries, with the abacus being one of the earliest examples.

The first electronic digital computer was developed in the 1940s.

The invention of the microprocessor in the 1970s revolutionized the computing industry.

Types of digital systems

Combinational logic circuits are digital circuits that produce a digital output based on one or more digital inputs.

Sequential logic circuits are circuits that use feedback to store state information.

Microprocessors are digital systems that are designed to perform a wide variety of tasks.

Advantages of digital systems

Digital systems offer higher accuracy, faster processing, more storage capacity, and easier duplication and distribution than analog systems.

Disadvantages of digital systems

Digital systems can be susceptible to hacking, viruses, and system failures. They also create a digital divide that can leave some people behind.

Thank you for allowing me to share the history and advancements of digital systems with you today!

Digital Technology: Changing the Way We Live

Introduction to Digital Technology

Digital technology has revolutionized the way we communicate, work, learn, and live. It has transformed numerous industries and created new opportunities for growth and innovation.

Advantages of Digital Technology

Digital technology makes our lives easier and more convenient.

It allows us to stay connected with people all over the world.

It can improve efficiency in many industries.

Disadvantages of Digital Technology

Overuse can lead to addiction and impact mental health.

Cyberbullying is a growing concern in the digital age.

Exposure to inappropriate content can be harmful, especially for children.

Role of Digital Technology in Business

Digital technology streamlines business operations, enhances communication, improves data analysis, and expands market reach. Embrace it to stay competitive.

Significance of Digital Technology in Education

Digital technology in education improves access to information, enhances communication, and increases engagement and collaboration among students and teachers. It also prepares students for the workforce.

Thank you for listening and exploring the role and significance of digital technology with me today!